

August 23, 2026

Dear HIVERY Hiring Team,

I am writing to apply for the Data Engineer - New Grad position with HIVERY. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python data-processing systems, backend APIs, machine learning workflows, and scalable research software, I am excited by the opportunity to contribute to HIVERY Curate and the data platform behind retail simulation and optimization.

What stands out to me about this role is that it is built for a new or recent graduate who can bring academic and professional engineering experience into a real product team. My graduate work required the same habits I would bring to HIVERY: writing maintainable code, debugging data workflows, communicating trade-offs clearly, and improving performance while staying careful about data quality. I like that the role combines feature development, data engineering standards, code review, and learning from senior engineers in a fully remote team.

As a Research Software Developer with BigGeo/Mitacs at the University of Calgary, I built C++ and Python systems for processing, storing, and retrieving large Earth observation datasets. I developed reproducible pipelines, automated satellite-imagery preprocessing, converted data into structured tensor files, and improved one encoding workflow from 233 seconds to 26 seconds using C++ multithreading. This work strengthened my ability to reason about data structures, scalability, performance, and reliability in high-volume data systems.

I also bring backend and database experience that connects well with product data engineering. As a Back-End Developer, I designed and implemented Django REST APIs backed by PostgreSQL, built authentication workflows, integrated payment-service logic, and maintained service and database behavior for NLP and speech products. Through teaching assistant roles at the University of Calgary, I mentored students in AWS-based data pipelines, cloud storage, large-scale data processing, SQL, debugging, and database coursework, which maps well to HIVERY's requirements around Python, cloud providers, data warehousing, ELT-style work, and big data foundations.

My machine learning background is another reason HIVERY interests me. I have built PyTorch workflows for digital elevation model super-resolution, evaluated models with terrain-aware metrics, and worked with data validation issues where apparently good results could hide weak generalization. I know this role is data engineering rather than model research, but that experience helps me understand how reliable pipelines, careful validation, and performance-aware engineering support downstream AI and optimization products.

I would bring a positive, self-managing, and communication-oriented approach to the team. My research and teaching work required clear written updates, careful documentation, and collaboration with technical and non-technical audiences. I am comfortable learning new technologies, asking precise questions, and taking ownership of small improvements while staying aligned with team standards.

Thank you for considering my application. I would be grateful for the opportunity to discuss how my Python, backend, data pipeline, cloud-data, and machine learning experience can contribute to HIVERY's Curate engineering team.

Sincerely,
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